PhD defences - Abstracts

Michał Bujak

TITLE: *Advances in Ride-pooling: Methods from Operations Research and Network Science*

Ride-pooling is an on-demand urban mobility service where travellers can be matched into shared rides. Compared to standard taxi (ride-hailing), travellers experience longer travel time (other passengers' pick-ups and drop-offs) and discomfort due to the sharing. Those negative effects are compensated with a lower fare expressed via a sharing discount. My PhD thesis is dedicated to two streams of theoretical analysis of the ride-pooling problem: network science and operations research.

Graph (network) structures, starting with the seminal paper by Santi et al., were the algorithmic backbone of the proposed algorithms. Various methods introduce ride-pooling travellers and drivers as nodes in simple and bipartite representations. Network algorithms were leveraged to find optimal results. Although networks were an intermediate tool, neither their emergence nor their properties had hitherto been studied. In the first article of this thesis, for the first time, we formalised four distinct network structures within ride-pooling. We introduce new, weighted representations that contain more information, based on stochastic analysis. To understand the role of the topologies of ride-pooling networks, we studied the correlation between a sharing discount (the control variable), network properties, and system performance indicators. Our analysis highlighted the existence of a critical mass embedded in network topology. This means that based on certain topological properties, we can assess the demand level required for the ride-pooling to become efficient.

During the pandemic, we observed that most ride and car sharing solutions were limited or halted as a result of both restrictions and travellers' risk awareness. We pondered whether we can effectively mitigate ride-pooling associated transmission risks while preserving efficacy of the system. As network representations are often applied to epidemic studies, we constructed a clustering model and applied it to ride-pooling networks.

In the model, people are confined to their clusters, effectively reducing transmission rates (measured with the epidemic threshold). Our algorithm, based on graph neural networks, successfully mitigated most of the infection risk while maintaining high system efficiency. We generalised our model to more applications. In our original method, the exact performance measure can be latent (even black-box) as long as it is approximated with network connectivity. By introducing a three-component loss function, we achieved low potential transmission, high system performance, and balanced cluster sizes. Beyond ride-pooling networks, we successfully applied our algorithm to trade and peer-to-peer networks and achieved promising results superior to both analytical and machine-learning baselines.

Farnoud Ghasemi

TITLE: *Agent-based Modelling of Ride-sourcing Dynamics*

The emergence of ride-sourcing markets with platforms such as Uber and Lyft has disrupted the urban mobility landscape. Understanding and projecting the complex dynamics of these markets require robust modelling and computational tools capable of capturing the non-linear interactions among diverse stakeholders—travellers, drivers, platforms, and regulators—that give rise to emergent system-level patterns.

While state-of-the-art models effectively estimate macroscopic equilibrium conditions, they fall short in capturing the underlying mechanism of growth that drives market evolution over time. In particular, microscopic models fail to reproduce the most essential growth phenomena in ride-sourcing markets: network effects, whereby the value created for users on one side increases as more users join either the same side (same-side network effects) or the opposite side (cross-side network effects). As a result, existing models remain inadequate for exploring and testing platform strategies or regulatory interventions.

This dissertation introduces and extends MoMaS, a two-sided Mobility Market Simulation framework, an agent-based modelling approach grounded in computational techniques, to provide an in-depth representation of evolution in ride-sourcing markets. The developed framework allows the implementation and evaluation of diverse platform strategies and regulatory interventions, capturing traveller and driver behaviour that adapts to market dynamics through novel S-shaped learning models. MoMaS is designed to capture the full spectrum of ride-sourcing market evolution, both at the aggregate system level and the individual agent level. It not only reproduces realistic platform growth trajectories by capturing network effects, but also reflects the diversity in agent behaviour by modelling individual profiles of heterogeneous travellers and drivers, ranging from reluctant to neutral to loyal participants.